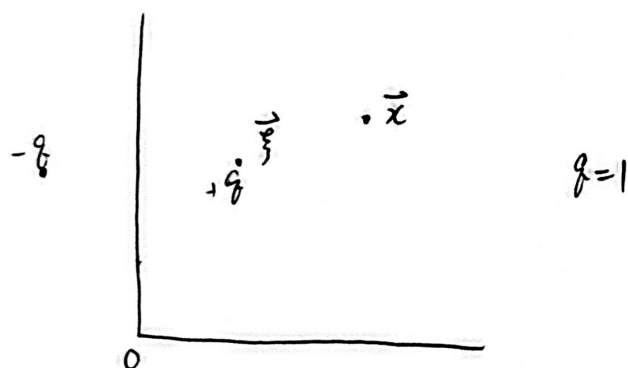


習題 2.22



$$G(\vec{x}, \vec{\xi}) = \frac{1}{2\pi} \left[\ln \frac{1}{|\vec{x} - \vec{\xi}|} + \ln \frac{1}{|\vec{x} + \vec{\xi}|} - \ln \frac{1}{|\vec{x} - \vec{\xi} + 2(\vec{\xi} \cdot \vec{j})\vec{j}|} - \ln \frac{1}{|\vec{x} - \vec{\xi} + 2(\vec{\xi} \cdot \vec{x})\vec{x}|} \right]$$

習題 2.23

$$\begin{cases} \partial_{tt} u - a^2 \partial_{xx} u = 0 \\ u(x, 0) = \varphi(x) \\ \partial_t u(x, 0) = \psi(x) \end{cases} \quad \text{現} \quad \begin{cases} \varphi(x) = \begin{cases} \frac{hx}{c}, & 0 \leq x \leq c \\ -\frac{h}{l-c}x + \frac{hl}{l-c}, & c < x \leq l \end{cases} \\ \psi(x) = 0 \end{cases} \quad (\text{旋轉前速度})$$

$$\begin{aligned} A_k &= \frac{2}{l} \int_0^l \varphi(\xi) \sin(k\pi\xi/l) d\xi = \frac{2}{l} \left[\int_0^c \frac{h\xi}{c} \sin(k\pi\xi/l) d\xi + \int_c^l \frac{h}{c-l} (\xi-l) \sin(k\pi\xi/l) d\xi \right] \\ &= \frac{2}{k\pi} \left[\frac{h}{c} \left(-c \cos\left(\frac{k\pi c}{l}\right) + \int_0^c \cos\left(\frac{k\pi\xi}{l}\right) d\xi \right) - \frac{h}{l-c} \left((c-l) \cos\left(\frac{k\pi c}{l}\right) + \int_c^l \cos\left(\frac{k\pi\xi}{l}\right) d\xi \right) \right] \\ &= \frac{2h}{k\pi} \cdot \frac{l}{k\pi} \left[\frac{1}{c} \sin\left(\frac{k\pi\xi}{l}\right) \Big|_0^c - \frac{1}{l-c} \sin\left(\frac{k\pi\xi}{l}\right) \Big|_c^l \right] \\ &= \frac{2lh}{(k\pi)^2} \cdot \sin\left(\frac{k\pi c}{l}\right) \cdot \frac{l}{c(l-c)} = \frac{2l^2h}{(k\pi)^2 c(l-c)} \sin\left(\frac{k\pi c}{l}\right) \end{aligned}$$

$$B_k = 0$$

$$\Rightarrow \text{~~the solution~~} \quad u(x, t) = \sum_{k=1}^{\infty} \frac{2l^2h}{(k\pi)^2 c(l-c)} \sin\left(\frac{k\pi c}{l}\right) \sin\left(\frac{k\pi x}{l}\right) \cos\left(\frac{k\pi at}{l}\right)$$

$$u(x, t) = \sum_{k=1}^{\infty} A_k \sin(k\pi x/l) e^{-k^2 \pi^2 a^2 t/l^2}$$

$$u(x, T) = \sum_{k=1}^{\infty} A_k e^{-k^2 \pi^2 a^2 T/l^2} \sin(k\pi x/l) = \varphi(x) = \sum_{k=1}^{\infty} \frac{1}{k^2} \sin(k\pi x/l)$$

$$\Rightarrow A_k = \frac{e^{\frac{(k\pi a)^2 T}{l^2}}}{k^2}$$

$$u(x, t) = \sum_{k=1}^{\infty} \frac{e^{\frac{(k\pi a)^2 (T-t)}{l^2}}}{k^2} \sin\left(\frac{k\pi x}{l}\right)$$

習題 2.25

$$(1) \begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X(l) = 0 \end{cases}$$

解. 特徵值: 當 $\lambda < 0$ 時, 特徵值 $\sqrt{-\lambda}, -\sqrt{-\lambda}$

$$X(x) = C_1 e^{-\sqrt{-\lambda}x} + C_2 e^{\sqrt{-\lambda}x} \Rightarrow X(0) = C_1 + C_2 = 0$$

$$X'(l) = C_1(-\sqrt{-\lambda}e^{-\sqrt{-\lambda}l} - \sqrt{-\lambda}e^{\sqrt{-\lambda}l}) = 0$$

$$\Rightarrow e^{-\sqrt{-\lambda}l} + e^{\sqrt{-\lambda}l} = 0, \text{ 無實數解.}$$

當 $\lambda = 0$ 時, $X(x) = Ax + B$, $B = X(0) = 0$

$$A = X'(l) = 0$$

$$\Rightarrow X(x) = 0$$

當 $\lambda > 0$ 時, $X(x) = B \sin(\sqrt{\lambda}x)$, $X'(l) = B\sqrt{\lambda} \cos(\sqrt{\lambda}l) = 0$

$$\sqrt{\lambda}l = \frac{2k+1}{2}\pi \Rightarrow \lambda = \left[\frac{(2k+1)\pi}{2l}\right]^2 \quad (k \in \mathbb{Z})$$

$$\Rightarrow \lambda_k = \left[\frac{(2k+1)\pi}{2l}\right]^2, \quad k \in \mathbb{Z}^+$$

$$X_k(x) = B_k \sin\left(\frac{(2k+1)\pi}{2l}x\right), \quad B_k \in \mathbb{R}$$

$$X(x) = \sum_{k=1}^{\infty} B_k \sin\left(\frac{(2k+1)\pi}{2l}x\right), \quad B_k \in \mathbb{R}$$

$$(2) \begin{cases} X''(x) + \lambda X(x) = 0, \quad 0 < x < l \\ X'(0) = X'(l) = 0 \end{cases}$$

解. $X(x) = B \cos(\sqrt{\lambda}x)$, $\sqrt{\lambda} = \frac{k\pi}{l} \Leftrightarrow \lambda_k = \left(\frac{k\pi}{l}\right)^2, \quad k \in \mathbb{Z}^+$

$$X_k(x) = -\frac{B_k}{\sqrt{\lambda}} \cos(\sqrt{\lambda}x)$$

$$\Rightarrow X(x) = \sum_{k=1}^{\infty} \frac{B_k}{k\pi} \cos\left(\frac{k\pi}{l}x\right), \quad B_k \in \mathbb{R}$$

$$(3) \begin{cases} X''(x) + \lambda X(x) = 0 \\ X(0) = X'(l) + hX(l) = 0 \end{cases}$$

解. 若 $\lambda > 0$ 时, $X(x) = C_1 (e^{\sqrt{\lambda}x} - e^{-\sqrt{\lambda}x})$

$$0 = X'(l) + hX(l) = C_1 \sqrt{\lambda} (e^{\sqrt{\lambda}l} + e^{-\sqrt{\lambda}l}) + C_1 h (e^{\sqrt{\lambda}l} - e^{-\sqrt{\lambda}l}) = 0$$

若 $\lambda < 0$ 时, $X(x) = B_k \sin(\sqrt{\lambda}x)$

$$0 = X'(l) + hX(l) = B_k (\sqrt{\lambda} \cos(\sqrt{\lambda}l) + h \sin(\sqrt{\lambda}l))$$

$$\tan(\sqrt{\lambda}l) = -\frac{h}{\sqrt{\lambda}} \quad \text{①}$$

特征值 λ_k 是 ① 的正根.

$$X(x) = \sum_{k=1}^{+\infty} B_k \sin(\sqrt{\lambda_k}x), \quad B_k \in \mathbb{R}$$

习题 2.21

$$(1) \varphi(x) = \sin^2 \frac{\pi x}{l} = \frac{1}{2} - \frac{1}{2} \cos\left(\frac{2\pi x}{l}\right)$$

$$A_k = \frac{2}{l} \int_0^l \left(1 - \cos\left(\frac{2\pi \xi}{l}\right)\right) \sin\left(\frac{k\pi \xi}{l}\right) d\xi = \frac{2}{l} \int_0^l \sin\left(\frac{k\pi \xi}{l}\right) d\xi - \frac{2}{l} \int_0^l \cos\left(\frac{2\pi \xi}{l}\right) \sin\left(\frac{k\pi \xi}{l}\right) d\xi$$

$$= \frac{1}{k\pi} [1 - (-1)^k] - \frac{1}{2l} \int_0^l \left[\sin\left(\frac{(k+2)\pi \xi}{l}\right) + \sin\left(\frac{(k-2)\pi \xi}{l}\right) \right] d\xi$$

$$= \frac{1}{k\pi} [1 - (-1)^k] - \frac{1}{2} \cdot \frac{1}{(k+2)\pi} [1 - (-1)^{k+2}] - \frac{1}{2(k-2)\pi} [1 - (-1)^{k-2}]$$

$$= \left[\frac{1}{k\pi} - \frac{1}{(k^2-4)\pi} \right] [1 - (-1)^k]$$

$$= \frac{(k^2-k-4)[1 - (-1)^k]}{k(k^2-4)\pi}$$

$$B_k = \frac{2}{k\pi a} \int_0^l \xi(l-\xi) \sin\left(\frac{k\pi \xi}{l}\right) d\xi = \frac{2l}{(k\pi)^2} \left(+ \int_0^l \cos\left(\frac{k\pi \xi}{l}\right) (l-2\xi) d\xi \right)$$

$$= \frac{2l^2}{(k\pi)^2} \left[(-1)^k (-l) + 2 \int_0^l \sin\left(\frac{k\pi \xi}{l}\right) d\xi \right]$$

$$= \frac{2l^2}{(k\pi)^2} \left[(-1)^{k+1} l + \frac{2l}{k\pi} (1 - (-1)^k) \right] = \frac{2l^3}{(k\pi)^3} \left((-1)^{k+1} + \frac{2}{k\pi} (1 - (-1)^k) \right)$$

$$u(x,t) = \sum_{k=1}^{+\infty} \left(\frac{(k^2-k-4)[1 - (-1)^k]}{k(k^2-4)\pi} \cos\left(\frac{k\pi at}{l}\right) + \frac{2l^3}{(k\pi)^3} \left((-1)^{k+1} + \frac{2}{k\pi} (1 - (-1)^k) \right) \sin\left(\frac{k\pi at}{l}\right) \right) \sin\left(\frac{k\pi x}{l}\right)$$

$$(2) \quad X(x) T''(t) - a^2 X''(x) T(t) + X(x) T(t) = 0$$

$$X(x) [T''(t) + T(t)] = a^2 X''(x) T(t)$$

$$\frac{T''(t) + T(t)}{T(t)} = a^2 \frac{X''(x)}{X(x)}$$

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ T''(t) + (\lambda a^2 + 1) T(t) = 0 \end{cases}$$

$$\frac{\pi}{2} (\lambda > 0) \quad X(x) = A \cos(\sqrt{\lambda} x), \quad \sqrt{\lambda} = \frac{k\pi}{L} \Rightarrow \lambda_k = \left(\frac{k\pi}{L}\right)^2$$

$$T(t) = C_1 \cos\left(\sqrt{\left(\frac{k\pi}{L}\right)^2 + 1} t\right) + C_2 \sin\left(\sqrt{\left(\frac{k\pi}{L}\right)^2 + 1} t\right) \Rightarrow C_1 = T(0) = \frac{1}{A \cos(\sqrt{\lambda} x)} \quad \text{不合理}$$

$$\text{由于 } \lambda = 0 \dots \text{故 } X(x) = A_0$$

$$T(t) = A_1 \cos(t) + A_2 \sin t, \quad u(x, 0) = A_0 A_1 = 1 \Rightarrow A_1 = \frac{1}{A_0}$$

$$\partial_t u(x, 0) = A_0 A_2 = 1 \Rightarrow A_1 = A_2 = \frac{1}{A_0}$$

$$\Rightarrow u(x, t) = \cos t + \sin t$$

$$(3) \quad X(x) T'(t) - X''(x) T(t) = X(x) T(t)$$

$$\frac{T'(t) - T(t)}{T(t)} = \frac{X''(x)}{X(x)} \Rightarrow \begin{cases} X''(x) + \lambda X(x) = 0 \\ T'(t) + (\lambda - 1) T(t) = 0 \end{cases}$$

$$X_k(x) = b_k \sin\left(\frac{k\pi}{L} x\right), \quad \lambda_k = \left(\frac{k\pi}{L}\right)^2$$

$$T_k(t) = A_k e^{(1-\lambda_k)t} = A_k e^{\left(1 - \left(\frac{k\pi}{L}\right)^2\right)t}$$

$$u(x, t) = \sum_{k=1}^{\infty} A_k b_k \sin\left(\frac{k\pi}{L} x\right) e^{\left[1 - \left(\frac{k\pi}{L}\right)^2\right]t}$$

$$u(x, 0) = \sum_{k=1}^{\infty} A_k b_k \sin\left(\frac{k\pi}{L} x\right) = 1,$$

$$(4) \quad X(x) T'(t) = a^2 X''(x) T(t) \Rightarrow \frac{T'(t)}{T(t)} = a^2 \frac{X''(x)}{X(x)}$$

$$\begin{cases} X''(x) + \lambda X(x) = 0 \\ T'(t) + \lambda a^2 T(t) = 0 \end{cases} \Rightarrow \begin{cases} X_k(x) = A_k \cos(\sqrt{\lambda_k} x), \quad \lambda_k = \left(\frac{k\pi}{L}\right)^2 = k^2 \\ T_k(t) = B_k e^{-(a^2 k^2)t} \end{cases}$$

$$u(x, t) = \sum_{k=1}^{\infty} A_k b_k e^{-(a^2 k^2)t} \cos\left(\frac{k\pi}{L} x\right)$$